



# AstroMagic

Version 2.4 Build 2.4-0006 | Photoshop UXP Plugin | Companion to PiMagic for PixInsight

Paired with PiMagic v2.4 (build v2.4-0006)

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**AstroMagic** is the companion Photoshop plugin that organizes PiMagic outputs into properly structured layer groups, ready for creative post-processing. It bridges the gap between automated processing in PixInsight and artistic finishing in Photoshop.

*"From PixInsight to Photoshop in one click — AstroMagic organizes your layers so you can focus on the art."*

**Workflow:** PiMagic processes your data in PixInsight and generates output files. AstroMagic imports those files into Photoshop as organized layer groups, with actions for further enhancement.

## Key Features

- **Automatic Layer Organization** — Creates properly named layer groups for SHO/HOO/RGB/narrowband palettes. Each channel and stretch variant is placed in the correct group with appropriate blending modes.
- **Mosaic Assembly** — Blends multi-panel mosaics using PiMagic's coordinate handoff. Import the parent PiMagic output folder and AstroMagic reads `mosaic_magic.json`, uses the PiMagic-aligned panes, skips Photoshop Auto-Align when geometry is complete, and runs Auto-Blend / layer organization. **A mosaic is now produced in a single PiMagic run — there is no "two-cycle", no "two-pass", and no user-started second PiMagic run.** In that one run PiMagic plate-solves every pane, reprojects them onto one common full canvas with PixInsight's `MosaicByCoordinates` engine, and hands the already-aligned panes to AstroMagic; AstroMagic then only blends them in Photoshop. The division of labour is fixed: PiMagic is the geometry oracle (one plate-solve per pane, store the WCS, `MosaicByCoordinates` joins the panes onto the canvas) and AstroMagic / Photoshop is the compositing engine (it blends only and never re-aligns when the coordinate handoff is present). The reported handoff contract is `handoff_mode = "full_canvas_no_alpha_lzw"`, which tells AstroMagic the panes are pre-aligned so it can skip Auto-Align and feather only the seams.
- **Save for PiMagic** — Legacy/manual fallback export for older two-pass workflows. The current public mosaic workflow does not require a second PiMagic pass; PiMagic creates the coordinate-

aligned mosaic handoff in one run.

- **Recursive Folder Loading** — Import Folder scans all subfolders automatically. You can select the top-level PiMagic output folder and AstroMagic will find all TIFF files, metadata, and annotation overlays.
- **Auto-Endpoints** — Histogram isolation per channel for optimal starting point. Automatically sets black and white points to bring out detail without clipping.
- **Stars Hue/Saturation** — Presets for controlling star colors. Quickly adjust star hues, reduce oversaturated stars, or enhance natural star colors.
- **Mono-to-RGB Conversion** — Convert individual monochrome channel layers into color composites directly within Photoshop.
- **PsMagic Actions** — One-click Photoshop Actions for layer blending, star manipulation, narrowband palette switching, continuum subtraction, and structure enhancement.
- **AutoMode Hand-Off from PiMagic** — The AstroMagic side of a PiMagic → Photoshop hand-off. When PiMagic starts a handoff, it writes a `Monitor_start.json` signal into the AstroMagic UXP plugin data folder; the panel's background poller picks it up within a few seconds and begins watching the PiMagic output folder on a 5-second cadence. When PiMagic finishes, it writes `Complete.txt` and `_pimagic.json`. For mosaics, PiMagic also writes `mosaic_magic.json` and aligned pane TIFFs. AstroMagic imports the parent PiMagic folder, preserves metadata, skips Photoshop Auto-Align when the coordinate handoff is complete, runs Auto-Blend where needed, saves the PSB, and writes `Import_complete.json` back to the output folder.
- **Palette Visibility Rules** — Automatically manages layer visibility based on your selected narrowband palette. When using SHO, plain RGB layers are hidden. When using DNBHO, both RGB and narrowband layers remain visible.

## User Interface

AstroMagic's interface is organized into six tabs:

### Import Tab

The primary tab for loading files into Photoshop.

- **Import Folder** — Opens a folder picker. For PiMagic mosaics, select the parent `PiMagic` output folder so AstroMagic can read `mosaic_magic.json`, `_pimagic.json`, `Images/By_Pane/Align_by_Coordinates/`, and `Images/Complete_Mosaic/`. Directly selecting the aligned-pane folder can hide the sidecars from Photoshop's UXP sandbox.
- **Import Images** — Opens a file picker to select individual TIFF files for import.
- **Save Mosaics** — Legacy/manual fallback export for older or manually assembled mosaics. It is not required for the current PiMagic single-run mosaic workflow, where PiMagic aligns the panes by

coordinates (MosaicByCoordinates) onto one full canvas and hands them to AstroMagic to blend — all in one run, with no second PiMagic pass.

- **Narrowband Palette** — Select the color palette for narrowband data: SHO (Hubble Palette), HOO (H Red + O Cyan), or Ha Enhanced (Red Only).
- **Merge Options** — “Merge Base with End Points” merges the base layer with the auto-endpoints adjustment for a flattened starting point.

## Global Tab

- **Load Layers from Files** — Load individual files as layers into the current document.
- **Make Annotations Layer Group** — Creates an organized group for annotation, census, and sky map overlays.
- **Make Stars Layer Group** — Creates a stars layer group with Hue/Saturation and Levels adjustment layers.
- **Make Global Adjustments Layer Group** — Creates a group with Hue/Saturation, Color Balance, Levels, Burn and Dodge, and End Points adjustments.

## Filters Tab

Create individual filter layer groups for narrowband and broadband channels:

- **SHO Groups** — Sulfur (Red), Hydrogen (Green), Oxygen (Blue)
- **HOO Groups** — Hydrogen (Red), Oxygen (Cyan)
- **Broadband Groups** — Mono Luminance, Red, Green, Blue

Each group is created with appropriate adjustment layers (Hue/Saturation, Color Balance, Levels, Burn and Dodge, End Points) for immediate creative processing.

## Enhance Tab

Enhancement tools for fine-tuning your image after layer organization.

## Continuum Tab

Continuum subtraction tools for narrowband processing. Removes broadband continuum emission from narrowband channels for cleaner nebula data.

## Tools Tab

AstroMagic availability status, about information, and diagnostic tools. AstroMagic is free and does not require activation.

# Import Workflow

## Standard Import (Single Target)

1. Process your data in PiMagic (PixInsight). When complete, PiMagic saves output files to a dated folder.
2. In AstroMagic, click **Import Folder**.
3. Select the PiMagic output folder (e.g., `PiMagic-M82-2026-04-06/` ). AstroMagic will scan all subfolders automatically.
4. AstroMagic creates the Photoshop document with all layers organized into groups:
  - Starless groups with adjustment layers (one per filter/palette)
  - Stars group with color adjustment layers
  - Annotation overlay layers (Annotation, Star Census, Sky Map)
  - Image metadata text layer

## Layer Naming Convention

AstroMagic follows PiMagic’s standardized naming. Output files use the pattern:

| File Pattern                             | Layer Name              | Group                   |
|------------------------------------------|-------------------------|-------------------------|
| <code>Starless_ObjectName_SH0.tif</code> | Starless_ObjectName_SHO | Starless ObjectName SHO |
| <code>Starless_ObjectName_H00.tif</code> | Starless_ObjectName_HOO | Starless ObjectName HOO |
| <code>Starless_ObjectName_RGB.tif</code> | Starless_ObjectName_RGB | Starless ObjectName RGB |
| <code>Stars_ObjectName_SH0.tif</code>    | Stars_ObjectName_SHO    | Stars                   |
| <code>Stars_ObjectName_Ha.tif</code>     | Stars_ObjectName_Ha     | Stars                   |
| <code>Final_ObjectName_RGB.tif</code>    | Final_ObjectName_RGB    | (base layer)            |
| <code>Annotation_Overlay_*.tif</code>    | Annotation              | Annotations             |
| <code>Census_Overlay_*.tif</code>        | Star Census             | Annotations             |
| <code>SkyMap_Overlay_*.tif</code>        | Sky Map                 | Annotations             |

## Palette Visibility Rules

When importing narrowband data, AstroMagic automatically manages layer visibility based on the selected palette:

- **SHO (Hubble):** SHO groups visible, plain RGB groups hidden. This prevents the broadband image from washing out the false-color narrowband palette.
- **HOO:** Both HOO and RGB groups remain visible.
- **DNBHO + RGB:** Both dualband narrowband and RGB groups remain visible.

You can always toggle layer visibility manually in the Layers panel after import.

## Mosaic Workflow

PiMagic and AstroMagic work together in a single-run mosaic workflow. PiMagic performs the coordinate alignment; AstroMagic performs blending and layer organization.

**One run, not two.** You no longer run PiMagic a second time to build a mosaic. There is no “two-cycle”, no “two-pass”, and no user-started second PiMagic pass. In a single run PiMagic stacks and processes the panes, plate-solves each pane and caches its WCS, then reprojects every pane onto one common full canvas with PixInsight’s `MosaicByCoordinates` engine (origin 0,0, no per-pane crop — each reprojected pane is the full canvas size and is zero outside that pane’s sky footprint). It writes those already-aligned per-pane TIFFs into `Images/By_Pane/Align_by_Coordinates/` plus a `mosaic_magic.json` sidecar that describes the canvas WCS and the aligned-pane paths. AstroMagic then imports those aligned panes and only *blends* them in Photoshop — it does not align them.

### Stage 1: Process and Align in PiMagic

1. Load your multi-panel data into PiMagic. Pane naming, folder structure, FITS `OBJECT` values, pointing clusters, and trailing numeric patterns can all trigger mosaic detection.
2. Click **Start** once. PiMagic processes each pane, solves the geometry, runs `MosaicByCoordinates / PerPaneMBC` alignment, and writes aligned pane TIFFs under `Images/By_Pane/Align_by_Coordinates/`.
3. PiMagic writes `mosaic_magic.json` and `_pimagic.json` in the parent PiMagic output folder. These sidecars carry the canvas, pane offsets, rotation/orientation contract, equipment metadata, target name, acquisition details, and output roles.

### Stage 2: Blend in AstroMagic

1. In AstroMagic, click **Import Folder** and select the parent `PiMagic` output folder, not only the aligned-pane subfolder.
2. AstroMagic reads `mosaic_magic.json`, loads the PiMagic-aligned pane TIFFs, places them according to PiMagic geometry, and skips Photoshop Auto-Align when `geometry_complete` is true.

3. AstroMagic runs Auto-Blend / layer organization, imports complete-mosaic overlays from `Images/Complete_Mosaic/` when present, and saves the PSB beside the PiMagic outputs.
4. Because the panes arrive pre-aligned, AstroMagic logs *"Step 2: Auto-Align SKIPPED — panes pre-aligned by PI"* and goes straight to the blend. It imports the aligned full-canvas TIFFs as layers and runs Photoshop's Auto-Blend (the proven panorama / `mergeAlignedLayers` path) to feather the seams, then applies auto-endpoints to set levels. **Stars** roles are composited with a black-safe Lighten / max blend so that no pane's stars are dropped, while **Starless** and **Final** roles are feathered through Auto-Blend. None of this is a second PiMagic run — it is the blend half of the same single-run handoff.

**Mosaic alignment ownership:** PiMagic owns plate solving, `MosaicByCoordinates` / `PerPaneMBC` alignment, rotation, pane placement, and the `mosaic_magic.json` handoff. AstroMagic owns Photoshop blending and layer preparation. Photoshop Auto-Align is only a fallback when PiMagic marks the handoff geometry incomplete because a pane could not be solved.

## Legacy / Manual Fallback Terms

Older logs and older manuals may mention `Mosaic_in.json` , `Mosaic_out.json` , `_mosaic_pimagic` , *Save Mosaics*, Photomerge, or a PiMagic second pass. These names describe the previous two-pass mosaic round-trip and remain useful only when diagnosing older runs or manually recovering a mosaic that was not completed through the current coordinate handoff.

- **Save Mosaics** can still export manually assembled mosaic layers into a `_mosaic_pimagic` folder for legacy recovery.
- **Photomerge / Photoshop Auto-Align** is not the normal PiMagic mosaic path. It is reserved for fallback when PiMagic cannot complete coordinate geometry for one or more panes.
- **Current public procedure:** one PiMagic run, then AstroMagic import/blend from the parent PiMagic output folder.

## Supported Mosaic Configurations

### Nightscape / Landscape Mosaics — Two Alignment Methods

For multi-pane nightscape and landscape mosaics, PiMagic offers two ways to align the panes, selected in PiMagic with the **Mosaic:** dropdown next to the Nightscape checkbox. Both are still a **single PiMagic run** — the choice only changes *who* aligns the panes, not the number of passes. AstroMagic follows whichever method PiMagic recorded in `mosaic_magic.json` .

- **Coordinates (default, recommended)** — PiMagic plate-solves each nightscape pane, caches its WCS, and reprojects the panes onto a common full canvas with `MosaicByCoordinates` (forced to a Stereographic projection to preserve the wide panorama geometry). The handoff is written with `geometry_complete = true` , so AstroMagic **skips Photoshop Auto-Align** and only blends the pre-aligned panes — the same idea as a deep-sky mosaic, giving sub-pixel registration

and PiMagic-owned whole-mosaic annotation. Use this whenever the panes plate-solve reliably (there is real sky / stars to solve).

- **Photoshop Align+Blend (legacy v2.0 behaviour)** — PiMagic deliberately skips `MosaicByCoordinates` and the coordinate handoff, saves the pane outputs, and marks the handoff `requires_auto_align = true` with `alignment_owner = "AstroMagic_Photoshop_AutoAlign"` . AstroMagic then runs Photoshop’s feature-based **Auto-Align** on all panes and **Auto-Blends** them, rebuilding the synthetic Stars layer with a black-safe Lighten blend. Use this for very wide / ultra-wide landscape panoramas, large foregrounds, or sparse / unsolvable star fields where plate-solving the individual panes is unreliable, so you let Photoshop stitch them.

**Tip:** Leave the method on **Coordinates** unless plate-solving the individual nightscape panes fails — only then switch to **Photoshop Align+Blend**. Either way it is one PiMagic run followed by one AstroMagic import / blend.

| Configuration                                   | Output Layers                                        |
|-------------------------------------------------|------------------------------------------------------|
| RGB broadband (e.g., 2-panel DSLR)              | 1 assembled RGB layer                                |
| Mono narrowband SHO (e.g., 2-panel Ha+OIII+SII) | Ha, OIII, SII, SHO (combined), HOO (combined) layers |
| OSC + dualband (e.g., 4-panel L-Enhance)        | DNBHO, RGB layers                                    |

**Tip:** For narrowband mosaics, PiMagic can provide individual filter layers (Ha, OIII, SII) and pre-combined palette images (HOO, SHO). AstroMagic keeps those roles separate so Stars, Starless, Final, and palette layers do not overwrite each other.

## Installation

AstroMagic is installed exclusively from PiMagic’s **Tools** tab in PixInsight. Both the plugin and actions are bundled with PiMagic — no separate download required.

### Prerequisites

- Adobe Photoshop 2023 (v24.0) or later
- Adobe Creative Cloud Desktop app installed and running
- PiMagic v2.4 installed in PixInsight

- For PiMagic processing before AstroMagic import: stand-alone RC-Astro CLI installed and checked in PiMagic. PiMagic v2.4 uses `rc-astro` for BlurXTerminator, NoiseXTerminator, and StarXTerminator instead of the PixInsight RC-Astro process modules.

## Step 1: Install the AstroMagic Plugin

1. Launch PiMagic in PixInsight (Script → PiMagic Studio → PiMagic)
2. Use the first-run Welcome Wizard, *Tools* → *Check Plugins*, or the **Tools** tab
3. In the AstroMagic section, click **Install Plugin**
4. The Adobe Creative Cloud installer will launch automatically
5. Accept the security warnings ("Install locally" → "OK")
6. Wait for the installation to complete
7. Restart Photoshop

## Step 2: Install PsMagic Actions

**Important:** Photoshop must be open before clicking Install Actions.

1. Open **Photoshop** (it must be running)
2. In PiMagic, click **Install Actions** from the Welcome Wizard, *Tools* → *Check Plugins*, or the **Tools** tab
3. PiMagic connects directly to Photoshop and runs the installer automatically — no file browsing needed
4. The installer removes any existing PsMagic Actions set and loads the current version
5. A success confirmation will appear when installation is complete

## Accessing AstroMagic

After installation, access AstroMagic in Photoshop:

### Plugins → AstroMagic

The AstroMagic panel will open and can be docked anywhere in your workspace.

## Verify PsMagic Actions

Check that **PsMagic Actions** appears in your Actions panel (Window → Actions). There should be exactly one PsMagic Actions set — the installer handles cleanup automatically.

## PsMagic Actions Reference

The PsMagic Actions panel provides one-click Photoshop Actions for common astrophotography tasks. These actions are installed through PiMagic's Welcome Wizard, plugin checker, or Tools tab and complement AstroMagic's layer organization workflow.

## Available Actions

| Category                     | Actions                          | Description                                                         |
|------------------------------|----------------------------------|---------------------------------------------------------------------|
| <b>Layer Blending</b>        | Blend modes, opacity presets     | Quick layer blend operations for combining channel data             |
| <b>Star Manipulation</b>     | Star reduction, star enhancement | Control star size and appearance in your images                     |
| <b>Narrowband Palettes</b>   | SHO, HOO, palette switching      | Switch between narrowband color palettes                            |
| <b>Continuum Subtraction</b> | Continuum subtract               | Remove continuum from narrowband channels for cleaner emission data |
| <b>Structure Enhancement</b> | Structure layers, Smart Filters  | Enhance fine detail and structure in nebulae and galaxies           |

## Using Actions

1. Open the Actions panel: **Window** → **Actions**
2. Expand the **PsMagic Actions** folder
3. Select the desired action
4. Click the Play button (▶) to run the action

## License Status

AstroMagic is now free and independent of PiMagic license activation. There is no AstroMagic license key, activation button, refresh step, or transfer workflow.

1. Install AstroMagic.
2. Open it in Photoshop.
3. Use the Tools tab to confirm the panel shows **Free and ready**.

**Availability:** AstroMagic can now be installed on its own and used without a PiMagic entitlement on the same machine.

## Updating AstroMagic

AstroMagic updates are delivered through PiMagic's repository and installed from the current bundled files. To update:

1. In PixInsight, go to **Resources** → **Updates** → **Check for Updates**
2. Install the PiMagic update (AstroMagic files are included)
3. In PiMagic's Welcome Wizard, plugin checker, or Tools tab, click **Install Plugin** to install the current bundled plugin
4. If the bundled PsMagic Action set changed, click **Install Actions** (this automatically removes old versions)
5. Restart Photoshop

**Note:** Do not uninstall the old version first – this may delete your preferences and settings.

## Troubleshooting

### Plugin Not Appearing in Plugins Menu

1. **Restart Photoshop** completely (close and reopen)
2. Verify the plugin is installed: Go to **Plugins** → **Manage Plugins**
3. Look for AstroMagic in the list and make sure it is **enabled**
4. Check that your Photoshop version is 2023 (v24.0) or later

### Plugin Installation Fails

1. Make sure the .ccx file is on the same drive as Photoshop
2. Use the **Open Folder** button in PiMagic's Tools tab to verify the .ccx file exists
3. Quit Creative Cloud Desktop completely, restart it, then try again

### Actions Not Installing

1. Ensure **Photoshop is open** before clicking Install Actions in PiMagic

2. If the automatic installer fails, PiMagic will show a fallback dialog with manual instructions
3. Manual fallback: In Photoshop, go to File → Scripts → Browse... and select `Install_PsMagic_Actions.jsx` from the AstroMagic folder

## macOS Security Warning

If you see “can’t be opened because it is from an unidentified developer”:

1. Go to **System Preferences** → **Security & Privacy** → **General**
2. Click **Open Anyway** next to the blocked plugin message

## Video Tutorials

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Comprehensive video tutorials are available at [pimagicstudio.com/about/](https://pimagicstudio.com/about/)

YouTube: [Utah Desert Remote Observatories](#)

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